Kernel: Python 3 (CoCalc)

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DATASET OVERVIEW

TITLE : Predicting Income Levels Using Demographic Patterns

SOURCE: Adult Income Dataset

DESCRIPTION: This dataset provides data on the income of a group of people older than 16 years of age. The objective of this dataset is to predict whether a person's annual income exceeds $50,000, and factors influencing the value.

In [3]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [4]:
```
df = pd.read_csv('adult_income.csv')
df
```

Out[4]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
0	17	Private	148522	11th	7	Never-married	occupation	Own-c
1	17	Private	93235	12th	8	Never-married	occupation	Own-c
2	17	Private	184924	9th	5	Never-married	occupation	Own-c
3	17	Private	116626	11th	7	Never-married	occupation	Own-c
4	17	Private	209949	11th	7	Never-married	occupation	Own-c
...	...	...	...	...	...	...	...	...
31942	90	?	175444	7th-8th	4	Separated	?	Not-in family
31943	90	Federal-gov	195433	HS-grad	9	Married-civ-spouse	Craft-repair	Husba
31944	90	Private	47929	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husba

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
31945	90	?	313986	HS-grad	9	Married-civ-spouse	?	Husba
31946	90	Private	313749	HS-grad	9	Widowed	Adm-clerical	Unmai

31947 rows × 12 columns

In [5]:
```
df.shape
```

Out[5]:
```
(31947, 12)
```

In [6]:
```
df.head(5)
```

Out[6]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationshi
0	17	Private	148522	11th	7	Never-married	occupation	Own-child
1	17	Private	93235	12th	8	Never-married	occupation	Own-child
2	17	Private	184924	9th	5	Never-married	occupation	Own-child
3	17	Private	116626	11th	7	Never-married	occupation	Own-child
4	17	Private	209949	11th	7	Never-married	occupation	Own-child

In [7]:
```
df.tail(5)
```

Out[7]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
31942	90	?	175444	7th-8th	4	Separated	?	Not-in family
31943	90	Federal-gov	195433	HS-grad	9	Married-civ-spouse	Craft-repair	Husba
31944	90	Private	47929	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husba
31945	90	?	313986	HS-grad	9	Married-civ-spouse	?	Husba
31946	90	Private	313749	HS-grad	9	Widowed	Adm-clerical	Unmai

In [8]:
```
df.columns
```

Out[8]:
```
Index(['age', 'workclass', 'fnlwgt', 'education', 'education.num',
       'marital.status', 'occupation', 'relationship', 'race', 'sex',
```

```
       'native.country', 'income'],
      dtype='object')
```

In [9]:
```
df.info()
# I do not think any columns necessarily need to be converted from one type to
another.
```

Out[9]:
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31947 entries, 0 to 31946
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             31947 non-null  int64
 1   workclass       31947 non-null  object
 2   fnlwgt          31947 non-null  int64
 3   education       31947 non-null  object
 4   education.num   31947 non-null  int64
 5   marital.status  31947 non-null  object
 6   occupation      31947 non-null  object
 7   relationship    31947 non-null  object
 8   race            31947 non-null  object
 9   sex             31947 non-null  object
 10  native.country  31947 non-null  object
 11  income          31947 non-null  object
dtypes: int64(3), object(9)
memory usage: 2.9+ MB
```

In [10]:
```
df.dtypes
```

Out[10]:
```
age                int64
workclass         object
fnlwgt             int64
education         object
education.num      int64
marital.status    object
occupation        object
relationship      object
race              object
sex               object
native.country    object
income            object
dtype: object
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
age               0
workclass         0
fnlwgt            0
education         0
education.num     0
marital.status    0
occupation        0
relationship      0
race              0
sex               0
native.country    0
```

```
income            0
dtype: int64
```

In [12]:
```
df.describe()
```

Out[12]:

	age	fnlwgt	education.num
count	31947.000000	3.194700e+04	31947.000000
mean	38.570695	1.897319e+05	10.071149
std	13.646280	1.057567e+05	2.561430
min	17.000000	1.228500e+04	1.000000
25%	28.000000	1.176275e+05	9.000000
50%	37.000000	1.783120e+05	10.000000
75%	48.000000	2.374535e+05	12.000000
max	90.000000	1.484705e+06	16.000000

The mean gives a central value of the data. For this dataset the mean for age is 38, which means majority of the people are around 38 years of age. The min is the smallest value in the dataset and max is the largest. The standard deviation (std) shows how spread out the data is from the mean. Low std means the values are close to the mean and high std means the values are more spread out.

Analyzing for this dataset, The average age is 38, with values ranging from 17 to 90, indicating a wide age range. The relatively high standard deviation suggests there is significant variability in ages within the dataset.

In [13]:
```
print(df['age'].mean())
```

Out[13]:
```
38.5706952139481
```

Handling missing values

In [14]:
```
df = pd.read_csv('adult_income.csv', na_values= '?')
df.isnull().sum()
```

Out[14]:
```
age                 0
workclass        1778
fnlwgt              0
education           0
education.num       0
marital.status      0
occupation       1785
relationship        0
race                0
sex                 0
native.country     25
income              0
dtype: int64
```

In [15]:
```
df.fillna("Unknown", inplace=True)
df
```

Out[15]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
0	17	Private	148522	11th	7	Never-married	occupation	Own-c

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
1	17	Private	93235	12th	8	Never-married	occupation	Own-c
2	17	Private	184924	9th	5	Never-married	occupation	Own-c
3	17	Private	116626	11th	7	Never-married	occupation	Own-c
4	17	Private	209949	11th	7	Never-married	occupation	Own-c
...	...	...	...	...	...	...	...	...
31942	90	Unknown	175444	7th-8th	4	Separated	Unknown	Not-in family
31943	90	Federal-gov	195433	HS-grad	9	Married-civ-spouse	Craft-repair	Husba
31944	90	Private	47929	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husba
31945	90	Unknown	313986	HS-grad	9	Married-civ-spouse	Unknown	Husba
31946	90	Private	313749	HS-grad	9	Widowed	Adm-clerical	Unmar

31947 rows × 12 columns

In [16]:
```
df['occupation'].value_counts()
```

Out[16]:
```
occupation
Craft-repair         3838
Prof-specialty       3732
Exec-managerial      3696
Adm-clerical         3595
Sales                3388
Other-service        3148
Machine-op-inspct    1902
Unknown              1785
Transport-moving     1500
occupation           1462
Handlers-cleaners    1317
Farming-fishing       957
Tech-support          861
Protective-serv       617
Priv-house-serv       141
Armed-Forces            8
Name: count, dtype: int64
```

In [17]:
```
df['occupation'] = df['occupation'].replace('occupation', np.nan)
```

In [18]:
```
df['occupation'] = df['occupation'].fillna('Unknown')
```

In [19]:
```
df['occupation'].value_counts()
```

Out[19]:
```
occupation
Craft-repair         3838
Prof-specialty       3732
Exec-managerial      3696
Adm-clerical         3595
Sales                3388
Unknown              3247
Other-service        3148
Machine-op-inspct    1902
Transport-moving     1500
Handlers-cleaners    1317
Farming-fishing       957
Tech-support          861
Protective-serv       617
Priv-house-serv       141
Armed-Forces            8
Name: count, dtype: int64
```

In [20]:
```
(df == '?').sum()
```

Out[20]:
```
age               0
workclass         0
fnlwgt            0
education         0
education.num     0
marital.status    0
occupation        0
relationship      0
race              0
sex               0
native.country    0
income            0
dtype: int64
```

In [21]:
```
df.isnull().sum()
```

Out[21]:
```
age               0
workclass         0
fnlwgt            0
education         0
education.num     0
marital.status    0
occupation        0
relationship      0
race              0
sex               0
native.country    0
income            0
dtype: int64
```

In [22]:
```
[df.duplicated()]
```

Out[22]:
```
[0        False
 1        False
 2        False
 3        False
 4        False
          ...
 31942    False
 31943    False
 31944    False
 31945    False
 31946    False
 Length: 31947, dtype: bool]
```

Workclass, occupation and native.country columns have missing data.

They can affect analysis in the following ways:

```
- if they are just dropped, it could change the whole data
- can reduce dataset size
- affect conclusions negatively
```

In [23]:
```
df['education'].unique()
```

Out[23]:
```
array(['11th', '12th', '9th', '10th', '5th-6th', 'Some-college',
       'HS-grad', '7th-8th', 'Masters', 'Bachelors', 'Assoc-voc',
       'Assoc-acdm', 'Preschool', '1st-4th', 'Doctorate', 'Prof-school'],
      dtype=object)
```

In [24]:
```
# Adding a new column based on existing columns

def education_level(ed):
    if ed in ['Preschool','1st-4th','5th-6th','7th-
8th','9th','10th','11th','12th','HS-grad']:
        return 'Low Education'
    elif ed in ['Some-college','Assoc-acdm','Assoc-voc']:
        return 'Medium Education'
    else:
        return 'High Education'

df['edu_income_group'] = df['education'].apply(education_level) # This new
column (edu_income_group) represents how educational background could affect
income. I grouped people with lower, education, medium education and higher
education together to see how it plays a role.
```

In [25]:
```
df.head(5)
```

Out[25]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationshi
0	17	Private	148522	11th	7	Never-married	Unknown	Own-child
1	17	Private	93235	12th	8	Never-married	Unknown	Own-child
2	17	Private	184924	9th	5	Never-married	Unknown	Own-child

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationshi
3	17	Private	116626	11th	7	Never-married	Unknown	Own-child
4	17	Private	209949	11th	7	Never-married	Unknown	Own-child

In [26]:

```
# To create filters to extract subsets of my data
high_edu_high_income = df[
    (df['edu_income_group'] == 'High Education') &
    (df['income'] == '>50K')
]
high_edu_high_income
```

Out[26]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
3089	22	Private	233955	Bachelors	13	Never-married	Prof-specialty	Not-in family
3090	22	Private	100345	Bachelors	13	Never-married	Craft-repair	Not-in family
3091	22	Private	195532	Bachelors	13	Never-married	Prof-specialty	Not-in family
3835	23	Private	102942	Bachelors	13	Never-married	Unknown	Own-c
3836	23	Self-emp-not-inc	258298	Bachelors	13	Never-married	Unknown	Own-c
...	...	...	...	...	...	...	...	...
31908	90	Private	87372	Prof-school	15	Married-civ-spouse	Prof-specialty	Husba
31910	90	Private	46786	Bachelors	13	Married-civ-spouse	Sales	Husba
31924	90	Private	51744	Masters	14	Never-married	Exec-managerial	Not-in family
31933	90	Private	88991	Bachelors	13	Married-civ-spouse	Exec-managerial	Wife
31939	90	Private	206667	Masters	14	Married-civ-spouse	Prof-specialty	Wife

3803 rows × 13 columns

In [27]:

```
low_edu_low_income = df[
    (df['edu_income_group'] == 'Low Education') &
    (df['income'] == '>50K')
]
low_edu_low_income
```

Out[27]:

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relatic
944	19	Unknown	200790	12th	8	Married-civ-spouse	Unknown	Other-relativ
1124	19	Private	323605	7th-8th	4	Never-married	Other-service	Not-in family
2663	21	Private	548303	HS-grad	9	Married-civ-spouse	Prof-specialty	Own-c
3087	22	Self-emp-not-inc	202920	HS-grad	9	Never-married	Unknown	Unmar
3329	22	State-gov	186634	12th	8	Never-married	Exec-managerial	Not-in family
...	...	...	...	...	...	...	...	...
31822	79	Unknown	163140	HS-grad	9	Married-civ-spouse	Unknown	Husba
31851	81	Private	177408	HS-grad	9	Married-civ-spouse	Unknown	Husba
31883	83	Self-emp-inc	240150	10th	6	Married-civ-spouse	Farming-fishing	Husba
31911	90	Private	175491	HS-grad	9	Married-civ-spouse	Craft-repair	Husba
31945	90	Unknown	313986	HS-grad	9	Married-civ-spouse	Unknown	Husba

1893 rows × 13 columns

In [28]:

```
# I created a subset which shows the amount of people with high education
earning more than 50K and also created a subset of people with low education
earning more than 50K
print("Number of people with high education earning >50K:",
high_edu_high_income.shape[0])
print("Number of people with low education earning >50K:",
low_edu_low_income.shape[0])
```

Out[28]:

```
Number of people with high education earning >50K: 3803
Number of people with low education earning >50K: 1893
```

The above analysis directly shows that educational background plays a big role in people earning more than 50K a year

To list the unique values and categories and which categorical variables seem important for analysis or grouping

In [29]:

```
cat_cols = df.select_dtypes(include='object').columns
print(cat_cols)
```

Out[29]:
```
Index(['workclass', 'education', 'marital.status', 'occupation',
       'relationship', 'race', 'sex', 'native.country', 'income',
       'edu_income_group'],
      dtype='object')
```

In [30]:
```
for col in cat_cols:
    print(f"Column: {col}")
    print(df[col].unique()) # to show unique values in each categorical column
```

Out[30]:
```
Column: workclass
['Private' 'Local-gov' 'Federal-gov' 'Unknown' 'Self-emp-not-inc'
 'Self-emp-inc' 'State-gov' 'Never-worked' 'Without-pay']
Column: education
['11th' '12th' '9th' '10th' '5th-6th' 'Some-college' 'HS-grad' '7th-8th'
 'Masters' 'Bachelors' 'Assoc-voc' 'Assoc-acdm' 'Preschool' '1st-4th'
 'Doctorate' 'Prof-school']
Column: marital.status
['Never-married' 'Married-civ-spouse' 'Divorced' 'Widowed' 'Separated'
 'Married-spouse-absent' 'Married-AF-spouse']
Column: occupation
['Unknown' 'Handlers-cleaners' 'Other-service' 'Adm-clerical' 'Sales'
 'Craft-repair' 'Farming-fishing' 'Prof-specialty' 'Tech-support'
 'Priv-house-serv' 'Machine-op-inspct' 'Protective-serv'
 'Transport-moving' 'Exec-managerial' 'Armed-Forces']
Column: relationship
['Own-child' 'Not-in-family' 'Other-relative' 'Husband' 'Unmarried' 'Wife']
Column: race
['White' 'Black' 'Amer-Indian-Eskimo' 'Asian-Pac-Islander' 'Other']
Column: sex
['Male' 'Female']
Column: native.country
['United-States' 'India' 'Mexico' 'Philippines' 'Puerto-Rico' 'Canada'
 'Trinadad&Tobago' 'Haiti' 'El-Salvador' 'Poland' 'England' 'Peru'
 'Dominican-Republic' 'Scotland' 'Germany' 'Columbia' 'Honduras' 'Jamaica'
 'Italy' 'Vietnam' 'South' 'Guatemala' 'Portugal' 'Thailand' 'Japan'
 'Laos' 'Nicaragua' 'Hong' 'Yugoslavia' 'Unknown' 'France' 'Ecuador'
 'Outlying-US(Guam-USVI-etc)' 'Taiwan' 'Cuba' 'Greece' 'Iran' 'China'
 'Ireland' 'Hungary' 'Cambodia' 'Holand-Netherlands']
Column: income
['<=50K' '>50K']
Column: edu_income_group
['Low Education' 'Medium Education' 'High Education']
```

Grouping and Aggregation

In [31]:
```
grouped = df.groupby('edu_income_group')
mean_age = grouped['age'].mean()
mean_age
```

Out[31]:
```
edu_income_group
High Education      40.841487
Low Education       38.807563
Medium Education    36.360350
Name: age, dtype: float64
```

From the grouped data, education level strongly influences income, with people in the high education group being older on average and more likely to earn >$50K. People in the medium education comes next, and lastly, people in the Low Education.

Initial Observations

One of the most interesting insights from the Adult Income dataset is how strongly many factors such as education level, gender, workclass, and even relationship correlate with income. My focus was on Education Level, by grouping individuals into Low, Medium, and High Education categories, it becomes clear that those with higher education not only tend to be older on average but are also much more likely to earn over 50K annually. Additionally, examining categorical variables like occupation and marital status will reveal clear patterns in earnings, and show which groups are more likely to achieve higher income. Overall, the dataset highlights how factors such as education, work experience, and job type collectively influence financial outcomes, offering valuable perspectives for understanding socioeconomic trends and guiding career or policy decisions.

I noticed some columns that had question marks, and in the occupation column, I see just 'occupation' as the job, which also does not make sense. I am yet to work on this to find out why it is in the dataset in that way.

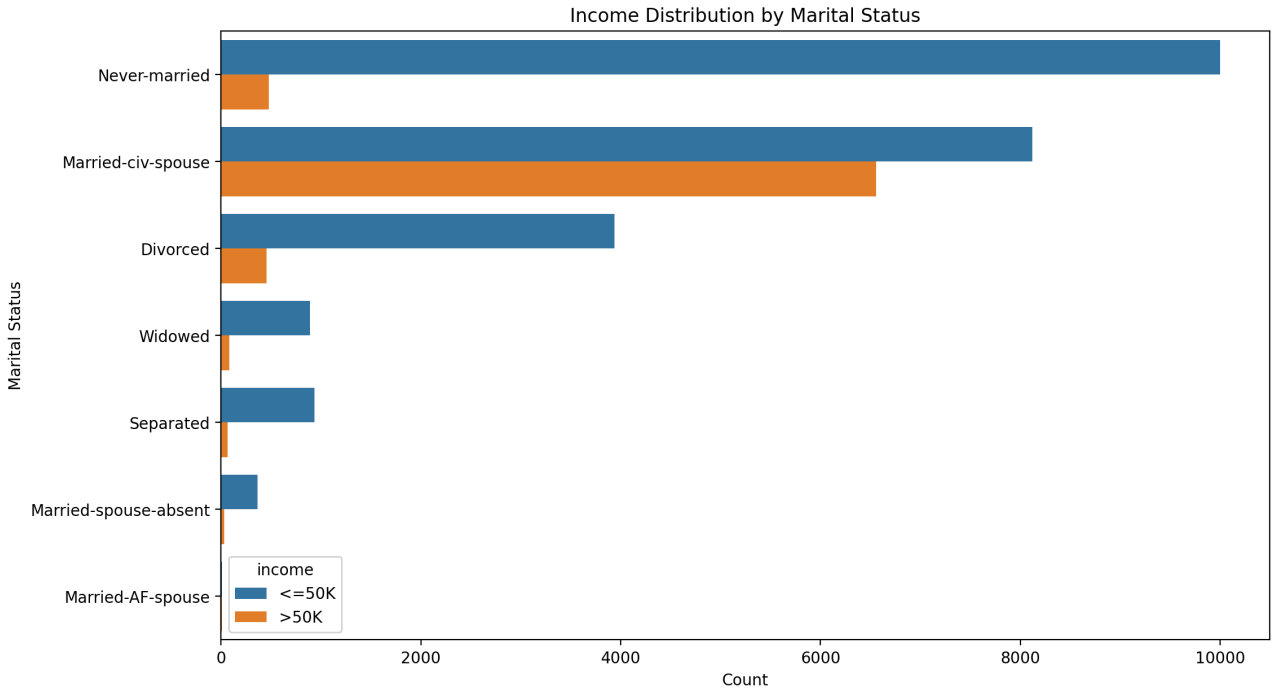
In [32]:
```
df.to_csv('adult_income_cleaned.csv', index=False)
```

In [33]:
```
df['occupation'].value_counts()
```

Out[33]:
```
occupation
Craft-repair         3838
Prof-specialty       3732
Exec-managerial      3696
Adm-clerical         3595
Sales                3388
Unknown              3247
Other-service        3148
Machine-op-inspct    1902
Transport-moving     1500
Handlers-cleaners    1317
Farming-fishing       957
Tech-support          861
Protective-serv       617
Priv-house-serv       141
Armed-Forces            8
Name: count, dtype: int64
```

In [34]:
```
sns.countplot(y='marital.status', hue='income', data=df)
plt.title("Income Distribution by Marital Status")
plt.xlabel("Count")
plt.ylabel("Marital Status")
plt.savefig("Income Distribution by Marital Status.jpg")
plt.show()
```

Out[34]:

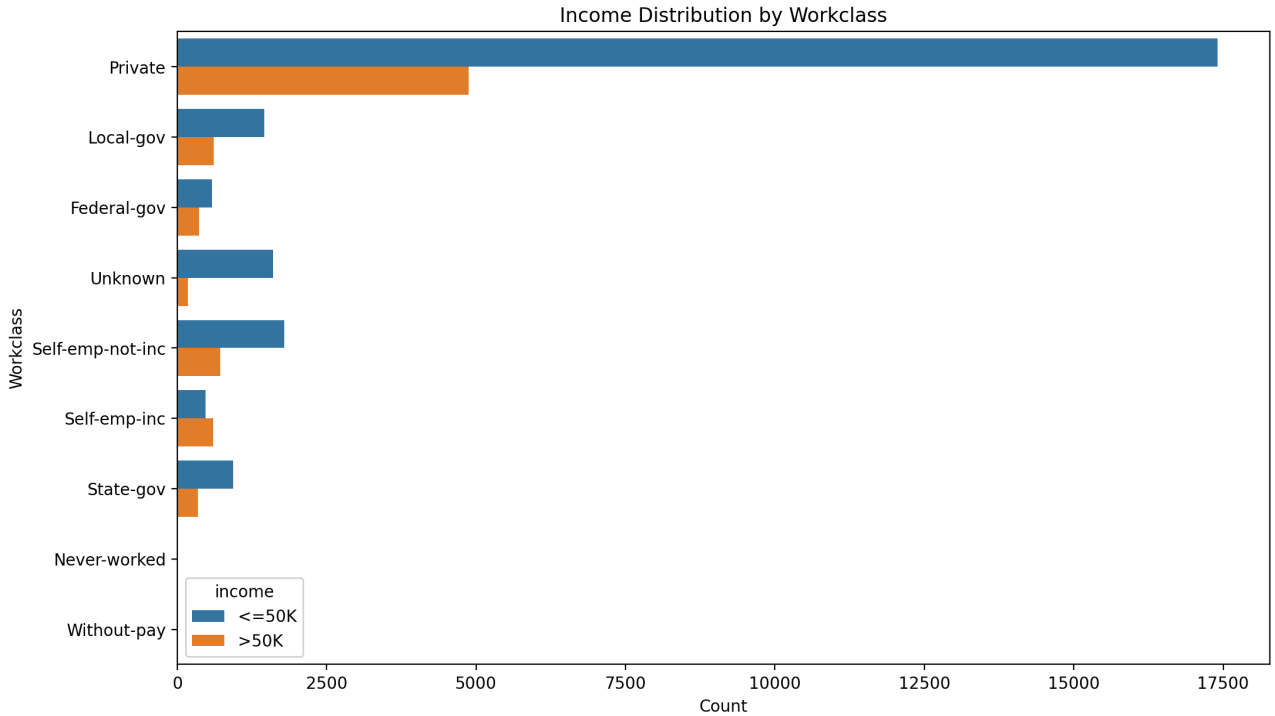


Never married individuals are more likely to earn above 50K

In [35]:

```
sns.countplot(y='workclass', hue='income', data=df)
plt.title("Income Distribution by Workclass")
plt.xlabel("Count")
plt.ylabel("Workclass")
plt.savefig("Income Distribution by Workclass.jpg")
plt.show()
```

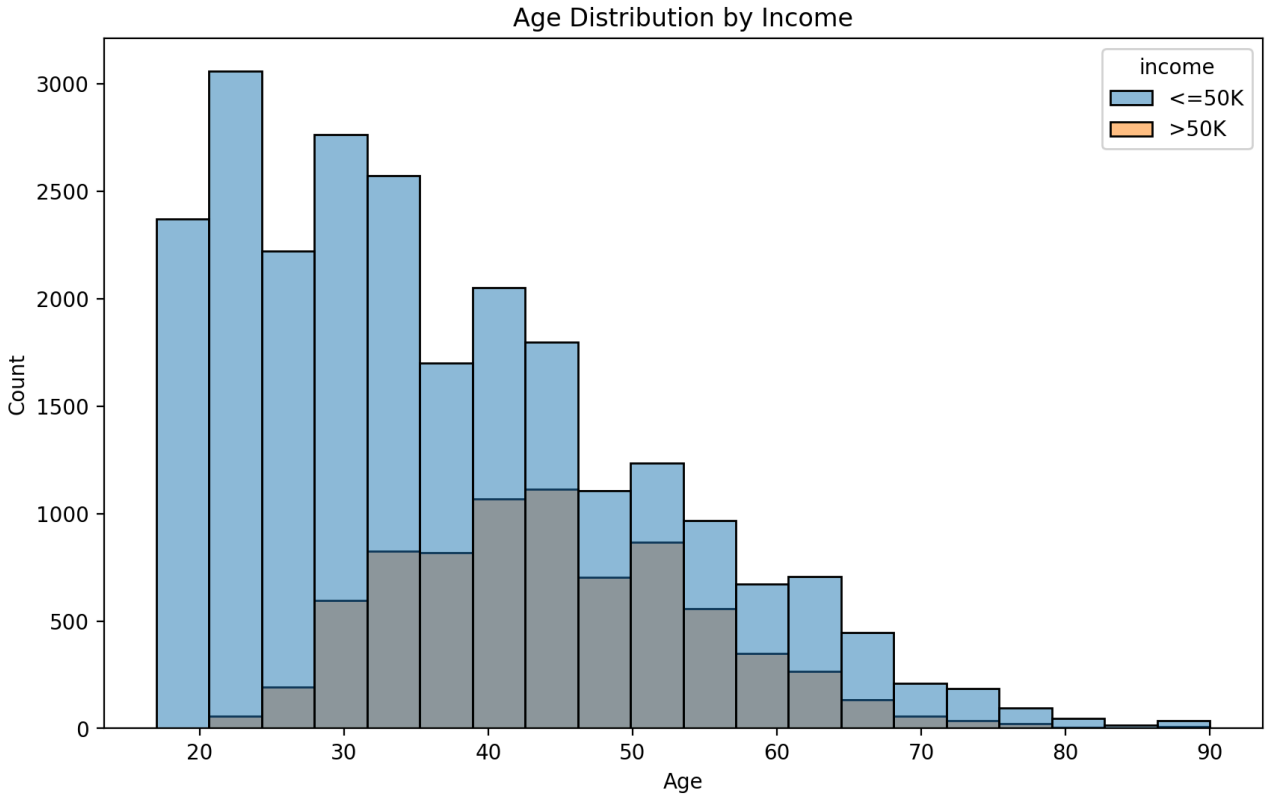
Out[35]:



People working in the private sector and self-employed roles tend to have a higher proportion of >50K earners compared to other work classes, while those with unknown ad other work classifications are more likely to earn ≤$50K.

In [36]:
```
plt.figure(figsize=(10,6))

sns.histplot(
    data=df,
    x='age',
    hue='income',
    bins=20
)

plt.title("Age Distribution by Income")
plt.xlabel("Age")
plt.ylabel("Count")
plt.savefig("Age Distribution by Income.jpg")
plt.show()
```

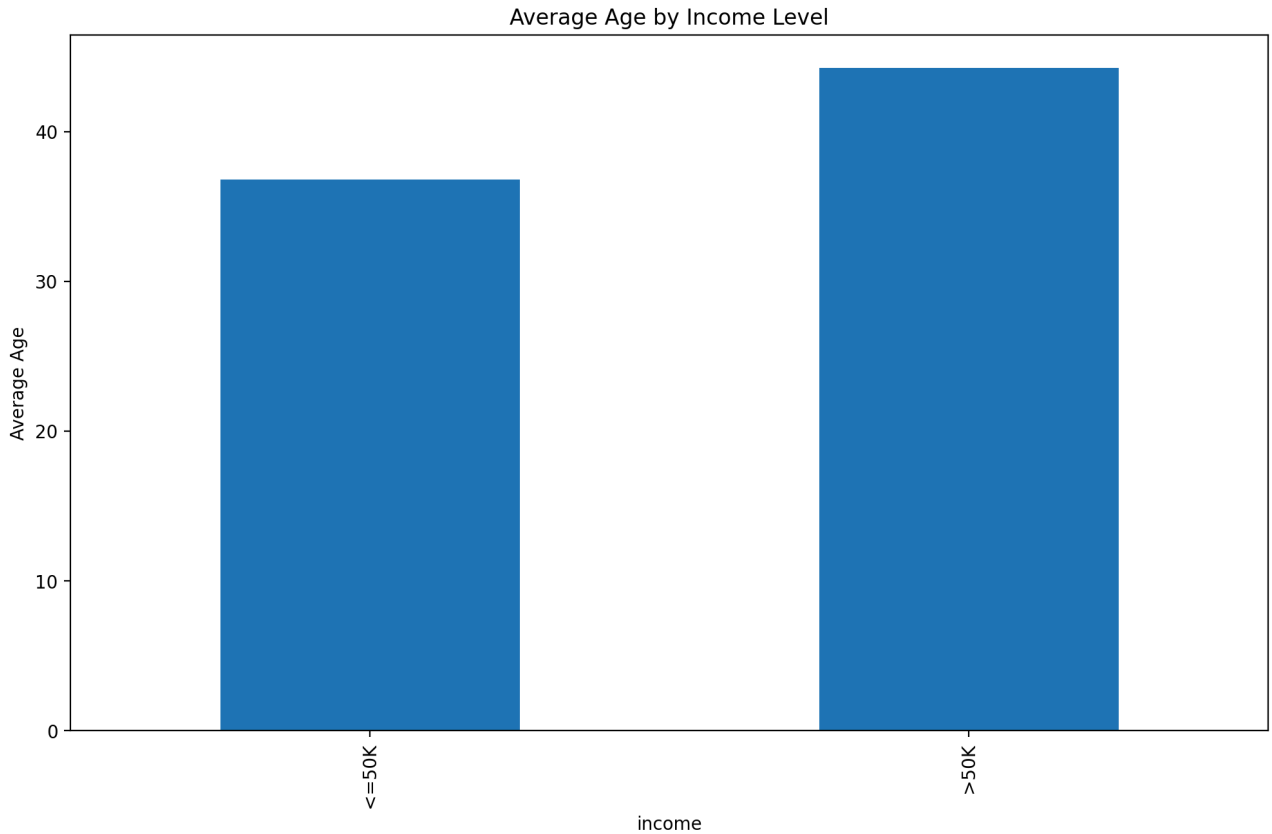
Out[36]:



The histogram shows how age values are distributed across income groups. The data is spread across a wide age range, but individuals earning above $50K are more concentrated in older age groups, while those earning less than or equal to $50K are more common among younger individuals. This suggests that age and experience plays a role in income differences.

In [37]:
```
avg_income_age = df.groupby('income')['age'].mean()

avg_income_age.plot(kind='bar')

plt.title("Average Age by Income Level")
plt.ylabel("Average Age")
plt.show()
```

Out[37]:

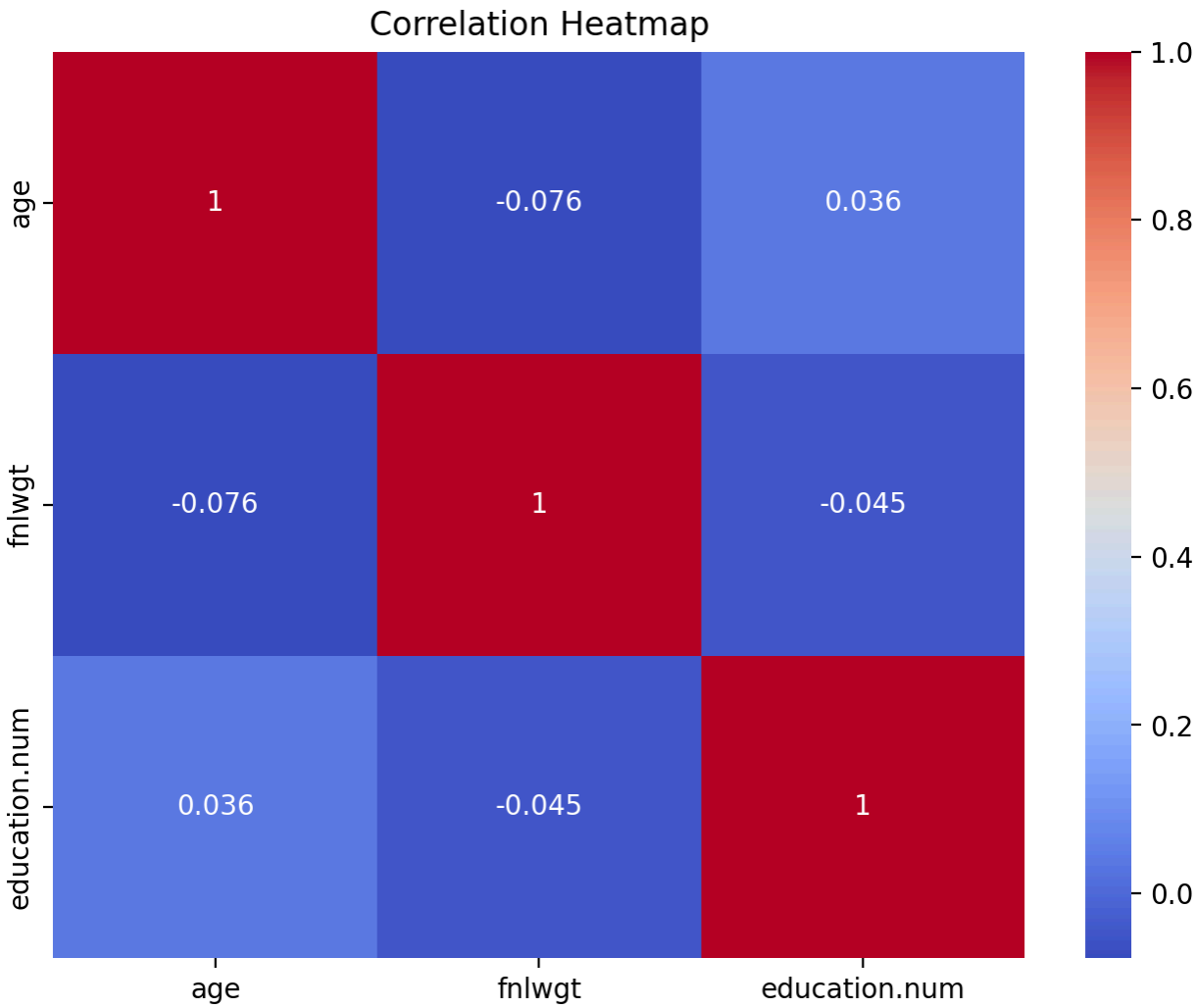


The results show that individuals earning above $50K have a higher average age compared to those earning less than or equal to $50K.

In [38]:

```
plt.figure(figsize=(8,6))

corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```

Out[38]:



The heatmap shows the correlation between numerical variables in the dataset. Most variables have weak correlations with each other, indicating that no single numerical factor strongly determines another.

In [39]:
```
df.to_csv('adult_income_cleaned.csv', index=False)
```

In [ ]: